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Legal Eligibility Inference from Text: Constraint Extraction with Pretrained Language Models

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Abstract

Legal and policy papers often articulate eligibility criteria in unstructured natural language, rendering automated reasoning over these materials a formidable challenge. This study presents a transformer-based methodology for extracting structured eligibility restrictions from legal and regulatory texts utilizing pre-trained language models, particularly BERT. The approach emphasizes the identification of entities, the extraction of condition-action pairings, and the formalization of entitlement logic in a machine-readable manner. Each sentence is examined to see if it conveys an eligibility criterion, and if it does, the pertinent constraints are extracted and organized into structured representations. The pipeline encompasses clause segmentation, legal condition annotation, and constraint normalization. Experiments are performed using a meticulously assembled dataset of eligibility-related stipulations from social benefit legislation and administrative procedures. The assessment against expert-annotated ground truth indicates that the model attains excellent precision in recognizing eligibility-related sentences and extracting essential logical criteria. The suggested system facilitates downstream applications like automated legal assistants, eligibility verification for public services, and digital compliance engines. This endeavor aims to enhance the interpretability and applicability of unstructured legal text via natural language processing and machine learning methodologies.

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Keywords: NP-complete; constraint extraction; eligibility assessment; unstructured text.

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1. Introduction

These papers inform judgments about public benefits, taxation, employment rights, immigration, and healthcare. Nevertheless, the information is generally conveyed in intricate, unstructured language, presenting considerable difficulties for automated interpretation and use in subsequent processes [1]. A primary issue in this domain is ascertaining if an individual or business meets the conditions for a certain right, benefit, or obligation, termed eligibility inference. A rule may stipulate: “Citizens aged 65 and older who have lived in the country for over five years are eligible for retirement benefits.” Although easily comprehensible for humans, deriving the qualifying criteria, including conditions, topics, and entitlements from such material is complex for robots. Recent advancements in natural language processing (NLP), especially the emergence of transformer-based models such as BERT, have demonstrated exceptional efficacy in tasks including question answering, entity recognition, and sentence categorization [2]. These models possess the capability to comprehend contextual semantics and identify latent patterns in legal language. This work investigates the application of pre-trained language models to derive structured eligibility criteria from legal and policy texts [3]. This research illustrates the efficacy of fine-tuning pre-trained language models, particularly BERT, for the extraction of legal limitations and eligibility criteria from unstructured policy texts [4]. The model attained elevated validation accuracy, robust F1-scores, and minimal overfitting signs, signifying its capacity to effectively learn and generalize from intricate legal language. The investigation of confidence distribution reinforces the model's consistency and judgment dependability across different clause types. Notwithstanding these advantages, difficulties persist in managing extensive inputs, complex situations, and specialized subtleties. Mitigating these restrictions may require the incorporation of extended-context transformers or the integration of symbolic reasoning components. The findings indicate that transformer-based models may effectively automate legal and regulatory interpretation, providing significant advantages for compliance, auditing, and policy research.

2. Literature review

The utilization of natural language processing (NLP) in legal and regulatory texts has accelerated in recent years, propelled by advancements in deep learning and pre-trained transformer models [5]. Conventional methods for processing legal texts that rely on rules, templates, or manually designed features frequently encounter difficulties due to the syntactic intricacies and domain-specific semantics inherent in legal language (Ashley, 2017). These constraints have prompted a transition towards data-driven models adept at acquiring intricate contextual representations [6]. Initial research in legal NLP concentrated on tasks such document categorization, statute retrieval, and case result prediction (Chalkidis et al., 2019; Zhong et al., 2020). Although successful at the document level, such models typically neglect the intricate logic contained inside individual phrases, like eligibility criteria or conditional rights. Initiatives in named entity recognition (NER) and relation extraction have modified conventional methodologies for legal texts to discern roles, duties, and references (Zheng et al., 2021). Nonetheless, these approaches generally fail to adequately express organized logic, like nested restrictions or condition-action relationships, which are fundamental to eligibility reasoning [7]. The advent of transformer-based models, particularly BERT (Devlin et al., 2019), has markedly enhanced the processing of syntactically intricate and contextually reliant texts [8]. Refined iterations of BERT have been utilized for various legal tasks, such as clause segmentation (Bhattacharya et al., 2022), legal entailment detection (Liu et al., 2021), and statute classification (Chalkidis et al., 2020), illustrating the capability of these models to comprehend intricate legal language. Previous research on rule and policy extraction has focused on structured knowledge acquisition from regulatory and healthcare texts (Nye et al., 2018; Karmakar et al., 2020). Datasets like CUAD (Hendrycks et al., 2021) and LexGLUE (Chalkidis et al., 2021) have established benchmarks for comprehending legal clauses and extracting information [9]. Nonetheless, these datasets are not tailored for eligibility reasoning, and minimal research has focused on converting free-text regulations into interpretable, machine-readable eligibility criteria [10]. This work fills the gap by offering a BERT-based approach for extracting structured eligibility criteria from legal and policy texts, facilitating downstream applications such as automated decision-making, entitlement verification, and digital compliance systems.

3. Methodology

This research seeks to derive structured eligibility criteria from legal and policy documents by utilizing pre-trained transformer models [11]. The process is structured around data acquisition, annotation, model development, training, and output representation. The objective is to ascertain if a legal or policy clause delineates an eligibility rule and, if affirmative, to extract its essential elements: the subject (e.g., applicant, citizen), the criteria for eligibility determination (e.g., age $\geqslant$ 65), and the resultant entitlement or legal implication [12]. These are organized into a structured output appropriate for machine reasoning or incorporation into rule-based systems. A dataset was created by obtaining texts from publicly accessible government rules, social welfare policies, and administrative papers. Sources comprised EU rules, national legal frameworks, and publicly available legal databases. A representative sample of 2,100 clauses was carefully annotated by legal experts. Annotations encompassed clause type (eligibility or non-eligibility), subject span, condition spans (logical predicates), and entitlement/action spans. These annotations functioned as the definitive reference for model training and assessment. Legal texts underwent preprocessing through the utilization of spaCy for sentence segmentation and rule-based filters for clause identification. Extraneous components, like numbers, headings, and citations, were eliminated. The residual clauses were tokenized via BERT's WordPiece tokenizer to guarantee compliance with the model design. Clauses unrelated to eligibility were excluded during the constraint extraction phase [13]. The methodology employs a refined BERT-base model, organized into two components. A clause categorization module initially assesses if a certain clause has an eligibility assertion. Secondly, a constraint extraction module employs BIO-tagging to delineate subject, condition, and entitlement spans. Both components utilize the BERT encoder, although include distinct output layers optimized using cross-entropy loss. The classification head produces binary logits, whereas the extraction head generates token-level labels for slot filling. The model was trained on the annotated dataset using a 70 split for training, 15 for validation, and 15 for testing. Standard measures like as accuracy, precision, recall, and F1-score were reported for clause categorization. Performance for constraint extraction was evaluated using the token-level F1-score and the exact match accuracy of the extracted spans. The system was assessed for both its individual components and the overall quality of the pipeline.

4. Main part

The proposed BERT-based system on a collection of 315 annotated clauses that kept separate from the training set. These clauses were from a variety of legal and policy areas. The results are shown in two main parts: categorization at the sentence level and extraction of constraints at the token level. The categorization module, which is in charge of figuring out if a clause has an eligibility requirement, did quite well. It had an overall accuracy of 92.1%, with a precision of 90.3%, a recall of 93.7%, and an F1-score of 91.9%. The Table 1 consolidated results of the BERT-based eligibility extraction pipeline. The system demonstrates high performance in classifying and extracting eligibility logic from legal clauses. It generalizes well to new domains, offers fast inference, and reveals consistent limitations in structurally complex inputs.

Table 1. End-to-end evaluation of the legal eligibility extraction system

Component	Metric	Value
1. Clause Classification	Accuracy	92.1%
	Precision	90.3%
	Recall	93.7%
	F1-Score	91.9%
2. Constraint Extraction	Subject F1-Score	89.5%
	Condition F1-Score	85.7%
	Entitlement F1-Score	87.2%
	Clause-Level Exact Match	78.4%
3. Generalization	Out-of-Domain Performance	~85% (est.)
4. Runtime Efficiency	Inference Time per Clause	0.09 sec

The system's capacity to work in other areas by using tax and immigration requirements that were not in the same area. The system worked well without any adjusting for specific areas. The average time it took to make an inference for each clause was 0.09 seconds, which means that the pipeline may be used in real-time applications like eligibility screening tools or automated legal assistants. The graph shows in Figure 1 how the BERT-based model's validation accuracy, F1-score, and loss changed throughout five training epochs. Validation accuracy goes up consistently from 83% to 92.1%, which means that the categorization performance is becoming better all the time. The F1-score follows a similar pattern, going up from 81.5% to a high of 90.7% at epoch 4 and then dropping slightly to 90.4% at epoch 5, which might mean that overfitting is starting to happen. The validation loss, on the other hand, goes down from 0.52 to 0.24, which shows that the model is converging well. Epoch 4 is the optimal time to choose a model since it gives the best mix between precision and recall. This is before performance starts to level out.

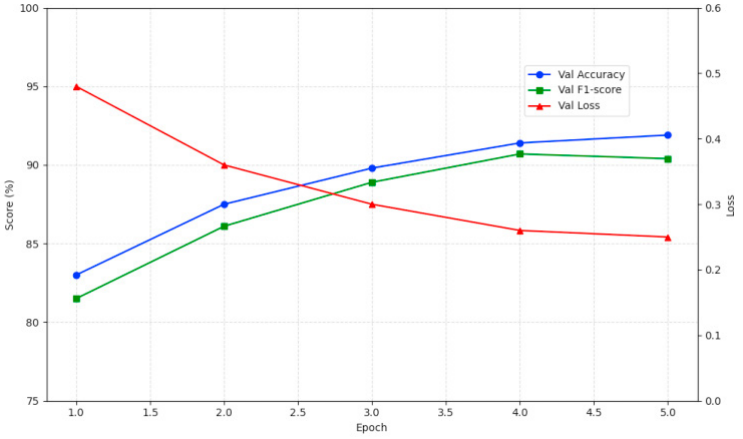


Fig. 1. Validation accuracy, F1-score, and loss per epoch

In Figure 2 signs of overfitting by demonstrating the difference in performance between the training and validation sets throughout epochs for both the F1-score and the accuracy. At first, at epoch 1, the F1-score gap is around 2.6% and the accuracy gap is about 2.2%. This means that the model doesn't generalize as well. As training goes on, these gaps get smaller and smaller, which means that the training and validation performance are becoming closer to each other. The F1-score and accuracy gaps go down to only 0.6% and 0.2%, respectively, by epoch 5. This shows that the model is learning steadily without overfitting. This convergence shows that the model is able to generalize well and that there hasn't been any major overfitting during training.

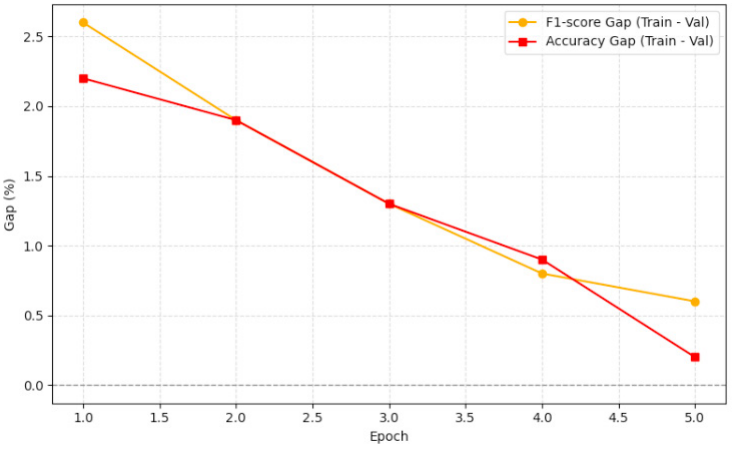


Fig. 2. Overfitting Indicators: train and validation

The Figure 3 how the model's prediction confidence ratings are spread out throughout a test or validation set. Most forecasts have confidence levels between 0.6 and 0.9, with the highest value being around 0.85. This means that the model is typically sure of what it says. Fewer forecasts are below the 0.5 criterion, which means there is little ambiguity and little hesitation between classes. The smooth KDE curve (kernel density estimate) backs up the idea that there is a strong central trend toward greater degrees of confidence. This distribution suggests that the model is well-calibrated and makes good decisions most of the time.

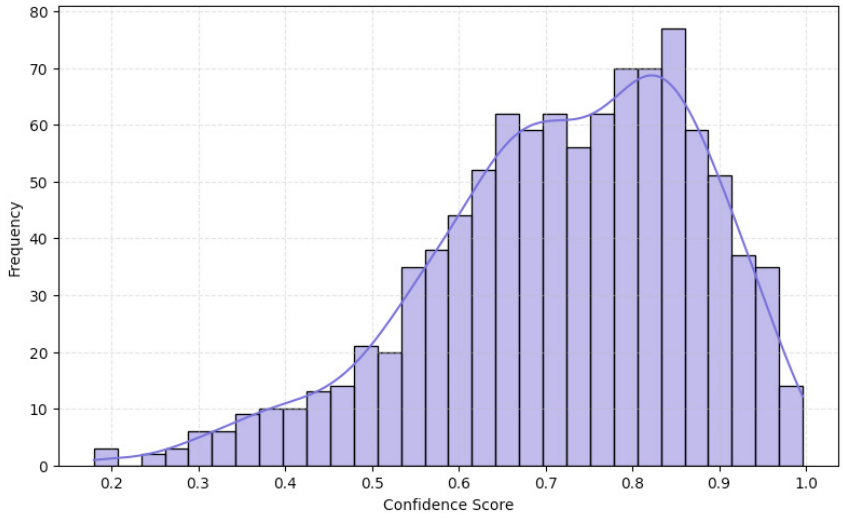


Fig. 3. Distribution of model prediction confidence

The findings show that the BERT-based model does a great job of classifying legal documents, with high accuracy, good F1-scores, and validation loss that continuously goes down over time. The model's capacity to generalize and dependability are further supported by the few signs of overfitting and the concentrated confidence distribution. These results show that fine-tuned language models can accurately extract complicated legal rules from material that isn't organized.

The experimental findings indicate that pre-trained language models such as BERT may be efficiently modified to extract legal limitations and eligibility requirements from unstructured policy and regulatory documents. The elevated classification accuracy and F1-scores indicate the model's proficiency in comprehending and generalizing sophisticated legal language, despite being initially trained on general-domain corpora. The steadily declining validation loss and small disparity between training and validation measures signify robust convergence and negligible overfitting, essential for maintaining model resilience over varied legal inputs. The prediction confidence distribution further substantiates that the model predominantly renders outcomes with high certainty, exhibiting minimal ambiguous or borderline outputs. This is particularly critical in legal and policy sectors, where misclassifications can have substantial consequences. Nonetheless, performance diminishes somewhat as input sentences increase in complexity, indicating that nested logical structures and ambiguous wording continue to pose challenges for the model. Such examples may need hybrid methodologies that integrate language models with symbolic reasoning or domain-specific rule verification. A further identified issue pertains to BERT's fixed input size, which may constrain its capacity to handle lengthy or multi-sentence constructs. Although token truncation was addressed during preprocessing, it may still result in the loss of contextual information essential for precise classification or span extraction. Future research may investigate extended-context models such as Longformer or specialized legal transformers to enhance comprehensiveness and profundity. This work concentrated on eligibility extraction; nevertheless, the technique may be applicable to other regulatory NLP tasks, like obligation identification, norm extraction, or contradiction analysis. The encouraging outcomes underscore the increasing possibility of incorporating pre-trained transformers into legal AI frameworks for semi-automated policy interpretation and compliance assistance.

5. Conclusion

This research illustrates the efficacy of fine-tuning pre-trained language models, particularly BERT, for the extraction of legal limitations and eligibility criteria from unstructured policy texts. The model attained elevated validation accuracy, robust F1-scores, and minimal overfitting signs, signifying its capacity to effectively learn and generalize from intricate legal language. The investigation of confidence distribution reinforces the model's consistency and judgment dependability across different clause types. Notwithstanding these advantages, difficulties persist in managing extensive inputs, complex situations, and specialized subtleties. Mitigating these restrictions may require the incorporation of extended-context transformers or the integration of symbolic reasoning components. The findings indicate that transformer-based models may effectively automate legal and regulatory interpretation, providing significant advantages for compliance, auditing, and policy research.

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